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Subject Code: BMB205

Roll No:

MBA

(SEM. II) THEORY EXAMINATION 2025-26

OPERATIONS MANAGEMENT

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

2x7 = 14

(a)	Define <i>work measurement</i> . State its one purpose.	CO1	K2
(b)	State any four characteristics of services that distinguish them from physical products.	CO1	K2
(c)	What is <i>service blueprinting</i> ? List its two key components.	CO1	K2
(d)	Define <i>aggregate production planning</i> . State its two objectives.	CO1	K2
(e)	What is <i>VED analysis</i> ? Give one example of a Vital spare part in a hospital.	CO1	K2
(f)	Define <i>Bullwhip effect</i> in supply chain management.	CO1	K2
(g)	What is <i>KAIZEN</i> ? State its core philosophy.	CO1	K2

SECTION B

2. Attempt any *three* of the following:

7 x 3 = 21

(a)	Explain the concept of <i>productivity</i> . Differentiate between <i>production</i> and <i>productivity</i> with a numerical example. Classify four factors affecting productivity into <i>controllable</i> and <i>uncontrollable</i> . Suggest two managerial strategies to improve productivity in a labor-intensive industry.	CO1	K2
(b)	Explain the service quality gap model (Parasuraman et al.). List the five dimensions of service quality as per SERVQUAL. Describe how SERVQUAL is used to measure service quality with a hypothetical example of a bank .	CO1	K2
(c)	A manufacturing company uses 9,000 units of a component annually. Ordering cost is ₹150 per order. Holding cost is 25% of unit price. Unit price is ₹40. Calculate: (i) Economic Order Quantity (EOQ) (ii) Number of orders per year (iii) Total inventory cost at EOQ Also, explain ABC analysis with a logical example of an automobile spare parts inventory.	CO1	K2
(d)	Explain the Bullwhip effect in supply chain management with its causes and consequences . A retailer sees a 5% demand increase but orders 20% more from the distributor. Why does this happen? Suggest three strategies to reduce the Bullwhip effect.	CO1	K2
(e)	Explain the PDCA cycle (Deming cycle) with a logical example of reducing defect rate in a factory. Differentiate between KAIZEN and Six Sigma on the basis of approach, scope and results.	CO1	K2

P.T.O.

SECTION C

3. Attempt any *one* part of the following:

7 x 1 = 7

(a)	Discuss the nature and scope of production and operations management in a modern organization. Explain the types of manufacturing processes with one example each.	CO1	K2
(b)	Describe the factors influencing plant location with suitable examples. Compare process layout and product layout on the basis of product variety, production volume, material handling cost, and flexibility.	CO1	K2

4. Attempt any *one* part of the following:

7 x 1 = 7

(a)	Explain the service design process step by step. Differentiate between <i>product design</i> and <i>service design</i> . How does service blueprinting help in identifying potential failure points? Give an example of a bank's ATM service blueprint .	CO1	K2
(b)	Discuss the dimensions of quality in services (SERVQUAL). Explain service capacity planning strategies – <i>lead strategy</i> , <i>lag strategy</i> , and <i>chase strategy</i> with examples from the Indian service sector .	CO1	K2

5. Attempt any *one* part of the following:

7 x 1 = 7

(a)	Describe the process of production planning and control (PPC) with its three main stages: routing, scheduling and loading . A company has 6 jobs on 2 machines. Explain the logic of Johnson's rule for scheduling (without numerical). Why is scheduling critical in batch manufacturing?	CO1	K2
(b)	Explain FSN analysis and HML analysis as inventory control techniques. A hospital wants to manage its surgical inventory. Which technique would you recommend for: (i) life-saving drugs, (ii) surgical gloves, (iii) high-value implants? Justify each.	CO1	K2

6. Attempt any *one* part of the following:

7 x 1 = 7

(a)	Discuss the conceptual model of supply chain management with its key drivers . Explain the difference between core supply chain and reverse supply chain . How does IT (e.g., ERP, RFID) help in improving supply chain performance?	CO1	K2
(b)	A company has monthly demand for a product over 6 months as follows: Jan: 100, Feb: 110, Mar: 115, Apr: 120, May: 125, Jun: 130 Calculate: (i) 3-month simple moving average for April, May, June (ii) 3-month weighted moving average for April (weights: 0.5, 0.3, 0.2 for most recent to oldest) (iii) Exponential smoothing for April (take March forecast = 110, $\alpha = 0.2$) Also explain why demand forecasting is critical in supply chain management .	CO1	K2

7. Attempt any *one* part of the following:

7 x 1 = 7

(a)	Explain Total Quality Management (TQM) . Describe Juran's quality trilogy (quality planning, quality control, quality improvement) with a practical example. Differentiate between Quality Circles and 7 QC Tools .	CO1	K2
(b)	Explain Total Productive Maintenance (TPM) and its eight pillars . Describe 5S (Seiri, Seiton, Seiso, Seiketsu, Shitsuke) with one example for each from a factory floor. How does TPM contribute to productivity improvement?	CO1	K2